Software Requirements Specification (SRS)

Project: Task Manager Application

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Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements of the Task Manager Application. The application is intended to help users create, organize, prioritize, schedule, track, update, and complete tasks in a simple and reliable manner.

1.2 Scope

The Task Manager Application covers task creation, editing, deletion, assignment of priority and status, due-date management, categorization, search and filtering, reminders, progress tracking, and user account management. It may support synchronization across sessions and devices. The application does not include enterprise project-management functions beyond the task-management features defined here.

1.3 Audience

Developers, QA Engineers, UI/UX Designers, System Administrators, Project Managers, End Users, and Assessment Evaluators.

1.4 Definitions

Task – a unit of work to be completed; User – an authenticated application user; Priority – importance level assigned to a task; Status – current state of a task; Due Date – date/time by which a task should be completed; Reminder – notification associated with a task; UI – User Interface; API – Application Programming Interface; CRUD – Create, Read, Update and Delete.

2. Overall description

2.1 Product perspective

The Task Manager Application is a standalone or web/mobile-based productivity system. It provides a user interface for managing tasks and a backend data service for storing user accounts, tasks, categories, priorities, due dates, reminders, and task history. The application may use notification and authentication services where required.

2.2 Major product functions

• User registration and login

• Create, view, edit and delete tasks

• Assign task priority and status

• Set due dates and reminders

• Organize tasks using categories/tags

• Search and filter tasks

• Sort tasks by priority, due date or status

• Mark tasks as completed

• View task progress and history

• Receive task reminders/notifications

• Manage user profile and preferences

• Synchronize task data across sessions

2.3 User roles and characteristics

User role	Characteristics
User	Basic computer/mobile skills; manages personal tasks and expects a simple, responsive interface.
Administrator	Manages users, system configuration and application-level monitoring where administrator features are enabled.

2.4 Operating environment

The application shall operate in a supported web browser or mobile/desktop environment, with a backend application server and database where applicable. It shall require network connectivity for server-backed functions and shall handle temporary connectivity failures gracefully.

2.5 Constraints

The application shall protect user credentials and task data, enforce appropriate access control, maintain data consistency, and operate within the supported hardware, browser, database, and network environments.

3. External interface requirements

3.1 User interfaces

The primary UI shall provide dashboards, task lists, task-entry forms, task-detail views, filters, search controls, status indicators, priority indicators, due dates, and notification settings. Forms shall provide validation and understandable error messages.

3.2 Hardware interfaces

No specialized hardware is required. The application shall operate on standard computers, tablets, or smartphones with supported input devices and display hardware.

3.3 Software interfaces

• Application database for persistent task and user data

• Authentication service/module for user login and authorization

• Notification service for reminders where enabled

• Optional calendar or synchronization API where integrated

3.4 Communications

Client-server communication shall use secure HTTPS/TLS connections where network communication is required. The application shall handle request failures and temporary network unavailability without corrupting stored task data.

4. System features (detailed)

Each requirement includes an acceptance criterion and test case reference. IDs follow TMA-F-###.

4.1 User Authentication and Account Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria	Test case
TMA-F-001	The system shall allow a new user to register using valid required account information.	Functional	High	User	Valid registration creates a unique user account.	TC-Auth-01
TMA-F-002	The system shall authenticate registered users using valid credentials.	Functional	High	User	Valid credentials allow access to the user's task dashboard.	TC-Auth-02
TMA-F-003	The system shall reject invalid login credentials and display an appropriate error message.	Functional	High	Security	Invalid credentials do not create an authenticated session.	TC-Auth-03
TMA-F-004	The system shall allow an authenticated user to log out securely.	Functional	High	User	Logout terminates the session and protected pages cannot be accessed without re-authentication.	TC-Auth-04

4.2 Task Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria	Test case
TMA-F-010	The system shall allow an authenticated user to create	Functional	High	User	A valid task is saved and appears in	TC-Task-01

	a task with a title and optional description.				the user's task list.	
TMA-F-011	The system shall allow users to edit existing tasks.	Functional	High	User	Updated task information is saved and displayed correctly.	TC-Task-02
TMA-F-012	The system shall allow users to delete tasks after appropriate confirmation.	Functional	High	User	Confirmed deletion removes the task from active task views.	TC-Task-03
TMA-F-013	The system shall allow users to mark a task as completed or reopen it.	Functional	High	User	Task status changes correctly between active and completed states.	TC-Task-04
TMA-F-014	The system shall prevent users from accessing or modifying tasks belonging to another user.	Functional	High	Security	Cross-user task access is denied.	TC-Sec-01

4.3 Priority, Status, Category and Scheduling

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria	Test case
TMA-F-020	The system shall allow users to assign a priority level to a task.	Functional	High	User	Selected priority is saved and displayed with the task.	TC-Org-01
TMA-F-021	The system shall allow users to assign and update task status.	Functional	High	User	Task status is saved and reflected in task views.	TC-Org-02
TMA-F-022	The system shall allow users to create or select categories/tags for organizing tasks.	Functional	Medium	User	Tasks can be associated with selected categories/tags.	TC-Org-03

TMA-F-023	The system shall allow users to set or change a task due date.	Functional	High	User	Valid due date is stored and displayed correctly.	TC-Org-04
TMA-F-024	The system shall allow users to configure reminders for tasks with a due date or selected reminder time.	Functional	Medium	User	Configured reminder is stored and triggered at the selected time where notifications are available.	TC-Org-05

4.4 Search, Filtering and Progress

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria	Test case
TMA-F-030	The system shall allow users to search tasks by title or description keywords.	Functional	Medium	User	Search returns matching tasks and excludes non-matching tasks.	TC-Search-01
TMA-F-031	The system shall allow users to filter tasks by status, priority and category/tag.	Functional	Medium	User	Selected filters display only matching tasks.	TC-Search-02
TMA-F-032	The system shall allow users to sort tasks by due date, priority or status.	Functional	Medium	User	Selected sort order is applied correctly.	TC-Search-03
TMA-F-033	The system shall provide a summary of active, completed and overdue tasks.	Functional	Medium	User	Dashboard counts match the underlying task data.	TC-Progress-01

4.5 Notifications and Administration

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria	Test case
TMA-F-040	The system shall notify users about configured	Functional	Medium	User	A notification is generated for an eligible reminder.	TC-Notify-01

	task reminders where notifications are enabled.					
TMA-F-041	The system shall allow users to manage notification and task preferences.	Functional	Medium	User	Preference changes are saved and applied to subsequent notifications.	TC-Notify-02
TMA-F-042	The system shall provide authorized administrators with user/account management functions where administrator access is enabled.	Functional	Medium	Administrator	Only authorized administrators can access administrative functions.	TC-Admin-01

5. Non-functional requirements (detailed)

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement	Test case
TMA-NF-001	The system shall load the main task dashboard within 3 seconds for at least 95% of requests under normal load.	Performance	High	95th percentile response time ≤ 3 seconds.	TC-Perf-01
TMA-NF-002	The system shall provide 99.5% monthly availability excluding scheduled maintenance.	Reliability	High	Monthly uptime is ≥ 99.5%.	TC-Reliability-01
TMA-NF-003	The system shall protect passwords and sensitive user/task data using appropriate	Security	High	Security audit confirms protected credentials and data.	TC-Sec-02

	security controls and shall not store passwords in plaintext.				
TMA-NF-004	The system shall maintain data consistency so that confirmed task changes are not lost during normal operation.	Data Integrity	High	Repeated create/update/delete tests preserve expected database state.	TC-Integrity-01
TMA-NF-005	The user interface shall be responsive and usable on supported desktop and mobile screen sizes and provide accessible labels and controls where applicable.	Usability/Accessibility	Medium	UI review and accessibility checks pass on supported screen sizes.	TC-UX-01

5.1 Security

5.1.1 Security Objectives

• Protect user credentials and personal task data from unauthorized access or disclosure.

• Ensure that task creation, modification, deletion and completion operations are authorized and maintain data integrity.

• Provide accountability through appropriate authentication, authorization and security-relevant logging.

5.1.2 Security Requirements

Req ID	**Requirement (shall...)**	**Type**	**Priority**	**Acceptance criteria / Test case**
TMA-SR-001	The system shall use HTTPS/TLS for network communication between supported clients	Security	High	Network traffic is encrypted. / TC-Sec-03

	and backend services.			
TMA-SR-002	The system shall store passwords only as secure password hashes using an approved password-hashing mechanism.	Security	High	Database inspection confirms plaintext passwords are absent. / TC-Sec-02
TMA-SR-003	The system shall enforce authorization so users can access and modify only their permitted resources.	Security	High	Unauthorized resource requests are rejected. / TC-Sec-01
TMA-SR-004	The system shall protect session credentials/tokens against unauthorized reuse and invalidate them on logout or expiry.	Security	High	Logged-out/expired sessions cannot access protected functions. / TC-Sec-04
TMA-SR-005	The system shall record security-relevant events such as failed logins and administrative actions without recording passwords.	Security	High	Required events are logged and sensitive credentials are absent. / TC-Sec-05

6. Quality attributes & Acceptance tests

• All high-priority functional requirements shall be implemented and verified.

• No critical security or data-integrity failures shall remain open at acceptance.

• All requirements shall have corresponding test cases in the RTM.

• Acceptance testing shall cover Authentication, Task Management, Organization, Search/Filtering, Notifications, Performance, Security, Reliability and Accessibility.

• Task data shall remain consistent after create, update, completion, deletion and recovery scenarios.

7. UML Use-Case diagrams

Figure 1: User Use-Case Diagram

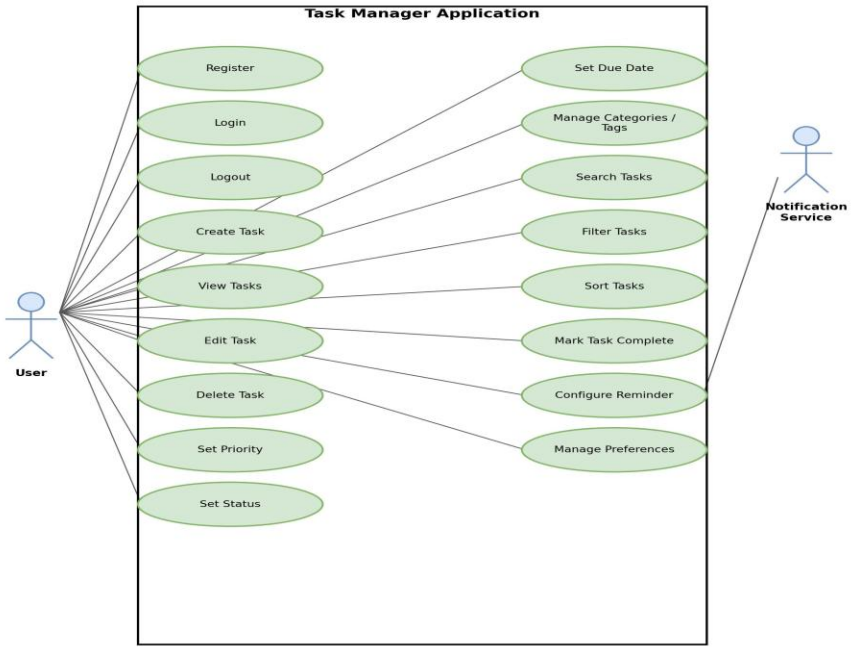
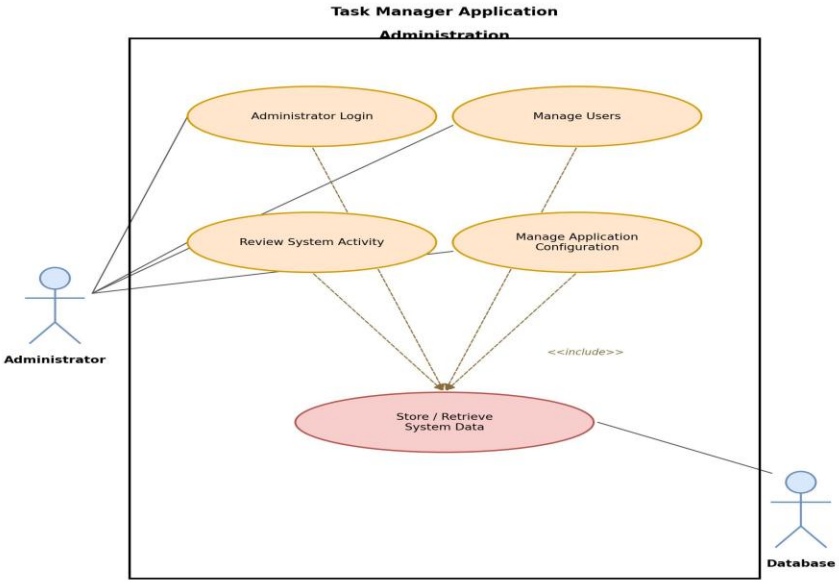


Figure 2: Administrator Use-Case Diagram



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)
TMA-F-001	User registration	4.1 / DS-Auth-01	Authentication Module	TC-Auth-01	N
TMA-F-002	User login	4.1 / DS-Auth-02	Authentication Module	TC-Auth-02	N
TMA-F-003	Invalid login rejection	4.1 / DS-Auth-03	Authentication Module	TC-Auth-03	N
TMA-F-004	User logout	4.1 / DS-Auth-04	Authentication Module	TC-Auth-04	N
TMA-F-010	Create task	4.2 / DS-Task-01	Task Module	TC-Task-01	N
TMA-F-011	Edit task	4.2 / DS-Task-02	Task Module	TC-Task-02	N
TMA-F-012	Delete task	4.2 / DS-Task-03	Task Module	TC-Task-03	N
TMA-F-013	Mark task complete/reopen	4.2 / DS-Task-04	Task Module	TC-Task-04	N
TMA-F-014	Task access authorization	4.2 / DS-Sec-01	Access Control	TC-Sec-01	N
TMA-F-020	Task priority	4.3 / DS-Org-01	Task Module	TC-Org-01	N
TMA-F-021	Task status update	4.3 / DS-Org-02	Task Module	TC-Org-02	N
TMA-F-022	Task categories/tags	4.3 / DS-Org-03	Task Module	TC-Org-03	N
TMA-F-023	Due date	4.3 / DS-Org-04	Scheduling Module	TC-Org-04	N
TMA-F-024	Task reminder	4.3 / DS-Org-05	Notification Module	TC-Org-05	N
TMA-F-030	Task search	4.4 / DS-Search-01	Search Module	TC-Search-01	N
TMA-F-031	Task filtering	4.4 / DS-Search-02	Search Module	TC-Search-02	N
TMA-F-032	Task sorting	4.4 / DS-Search-03	Search Module	TC-Search-03	N
TMA-F-033	Progress summary	4.4 / DS-Progress-01	Dashboard Module	TC-Progress-01	N
TMA-F-040	Notifications	4.5 / DS-Notify-01	Notification Module	TC-Notify-01	N
TMA-F-041	Notification preferences	4.5 / DS-Notify-02	Notification Module	TC-Notify-02	N
TMA-F-042	Admin user management	4.5 / DS-Admin-01	Administration Module	TC-Admin-01	N
TMA-NF-001	Dashboard response time	5 / DS-Perf-01	Application/UI	TC-Perf-01	N
TMA-NF-002	System availability	5 / DS-Reliability-01	Infrastructure/Ops	TC-Reliability-01	N

TMA-NF-003	Data security	5 / DS-Sec-01	Security Module	TC-Sec-02	N
TMA-NF-004	Data consistency	5 / DS-Integrity-01	Database/Task Module	TC-Integrity-01	N
TMA-NF-005	UI responsiveness/accessibility	5 / DS-UX-01	UI Module	TC-UX-01	N
TMA-SR-001	HTTPS/TLS	5.1.2 / DS-NetSec-01	Network Security	TC-Sec-03	N
TMA-SR-002	Password hashing	5.1.2 / DS-Sec-02	Authentication Module	TC-Sec-02	N
TMA-SR-003	Authorization	5.1.2 / DS-Access-01	Access Control	TC-Sec-01	N
TMA-SR-004	Session token protection	5.1.2 / DS-Sec-03	Session Management	TC-Sec-04	N
TMA-SR-005	Security event logging	5.1.2 / DS-Sec-04	Security/Audit Module	TC-Sec-05	N